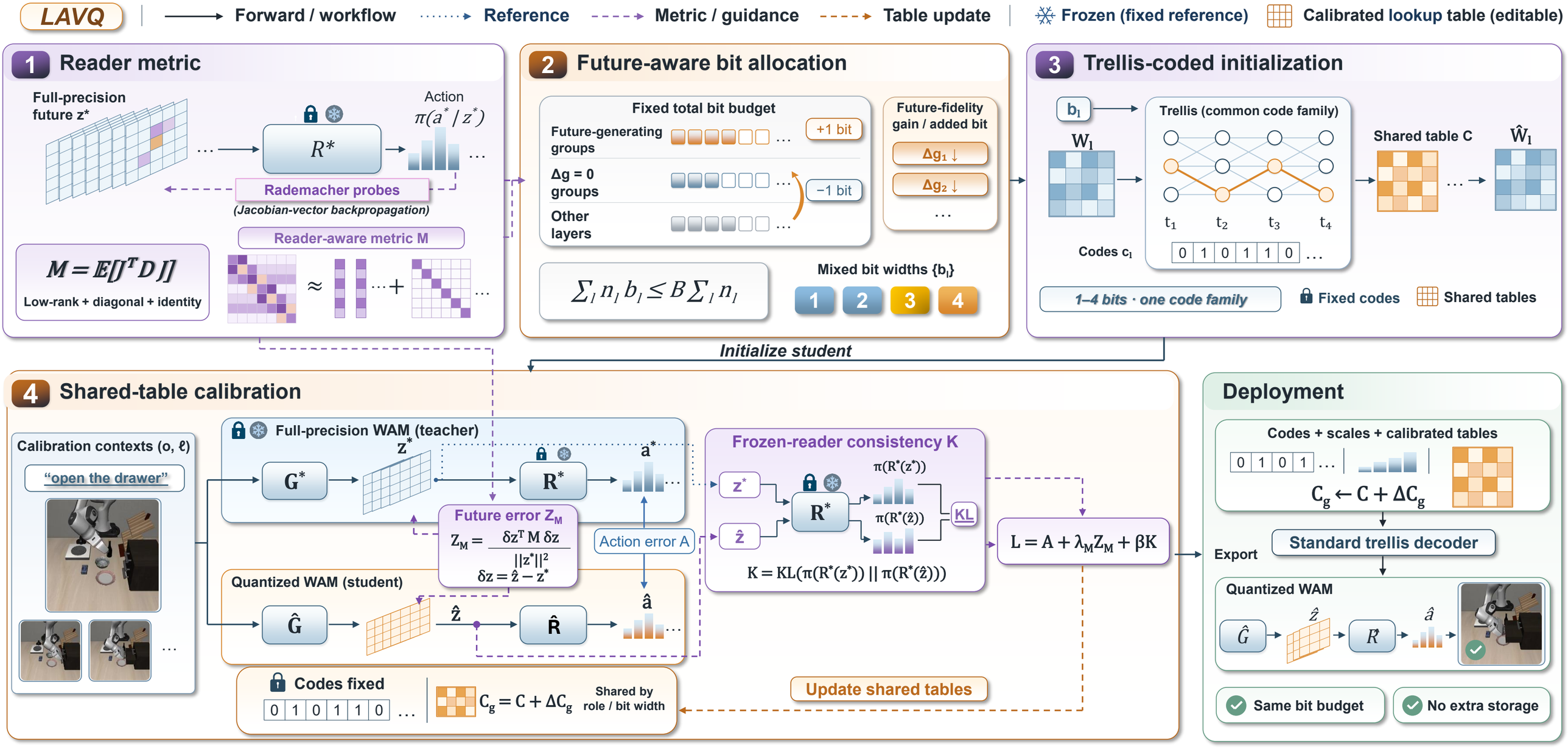

LAVQ
Forward / workflow
Reference
Metric / guidance
Table update
Frozen (fixed reference)
Calibrated lookup table (editable)
1 Reader metric
Full-precision future z*
Action
$\pi(a^*|z^*)$
R^*
Rademacher probes
(Jacobian-vector backpropagation)
Reader-aware metric M
$M = \mathbb{E}[J^T D J]$
Low-rank + diagonal + identity
2 Future-aware bit allocation
Fixed total bit budget
Future-generating groups
+1 bit
Δg = 0 groups
−1 bit
Other layers
Future-fidelity gain / added bit
$\Delta g_1 \downarrow$
$\Delta g_2 \downarrow$
$\sum_l n_l b_l \le B \sum_l n_l$
Mixed bit widths {b_l}
1
2
3
4
3 Trellis-coded initialization
b_l
W_l
Trellis (common code family)
t_1 t_2 t_3 t_4
Codes c_l
0 1 0 1 1 0 …
Shared table C
$\hat{W}_l$
1–4 bits · one code family
Fixed codes
Shared tables
Initialize student
4 Shared-table calibration
Calibration contexts (o, ℓ)
"open the drawer"
Full-precision WAM (teacher)
G^*
z^*
R^*
a^*
Future error Z_M
$Z_M = \frac{\delta z^T M \delta z}{||z^*||^2}$
$\delta z = \hat{z} - z^*$
Action error A
Quantized WAM (student)
$\hat{G}$
$\hat{z}$
$\hat{R}$
$\hat{a}$
Frozen-reader consistency K
$\pi(R^*(z^*))$
$\pi(R^*(\hat{z}))$
KL
$K = KL(\pi(R^*(z^*)) || \pi(R^*(\hat{z})))$
$L = A + \lambda_M Z_M + \beta K$
Codes fixed
0 1 0 1 1 0 …
$C_g = C + \Delta C_g$
Shared by role / bit width
Update shared tables
Deployment
Codes + scales + calibrated tables
0 1 0 1 …
$C_g \leftarrow C + \Delta C_g$
Export
Standard trellis decoder
Quantized WAM
$\hat{G}$ $\hat{z}$ R^* $\hat{a}$
Same bit budget
No extra storage